EE5203 L09 Worksheet

ADC and sensor acquisition - Basri Erdogan

Expected time: 30-45 minutes **Policy:** Attempt every question before consulting the worked solutions.

12-bit ADC, 3.3 V reference, `raw` in 0…4095 throughout.

Part A - Counts, volts, and resolution

1. Write the counts→millivolts formula, and state the LSB size in millivolts.
2. Complete the table using integer arithmetic only.

`raw`	mV	LED duty (`×1000/4095`)	Servo µs (`500 + ×2000/4095`)
0			
1024			
2867			
4095			

3. `mV = (raw / 4095) * 3300;` — explain exactly what this computes for `raw = 2000`, and give the corrected expression.
4. Would 16-bit resolution on the same 3.3 V reference help you distinguish two voltages 0.5 mV apart? Compute the LSB for both 12-bit and 16-bit and justify your answer — including one reason the extra bits might not help in practice.
5. A team wants to measure a 0–5 V sensor with this ADC. Explain what happens if they connect it directly, and describe a resistor divider that brings 5 V to just under 3.3 V. Give the two resistor values and the resulting `raw` for a 5 V input.

Part B - Configuration

6. Give the `MODER` bit pattern for an analog pin, and say what else changes inside the pin when analog mode is selected.
7. In one sentence each: what do **sampling time** and **conversion time** mean, and which one do you choose in CubeMX? Then calculate the conversion time for 84 cycles and for 3 cycles at an 8 MHz ADC clock.
8. Explain why a high-impedance source needs a longer sampling time. What symptom appears if it is too short *and you are switching channels*?
9. There is one ADC and sixteen channels. Describe what actually happens when `adc_read(ADC_CHANNEL_0)` is followed by `adc_read(ADC_CHANNEL_1)`, and state one consequence for a project that must correlate two sensors.
10. Contrast **single** and **continuous** conversion mode. Give one situation where each is the right choice.

Part C - Reading and flags

11. What sets `EOC`? Name the HAL call that waits for it, and say what clears the flag.
12. A loop calls `HAL_ADC_PollForConversion()` and `HAL_ADC_GetValue()` on every tick, but `HAL_ADC_Start()` only once, before the loop. Single conversion mode. Explain what `PollForConversion` returns from the second tick on, and what value the program reads.
13. Why is `HAL_ADC_PollForConversion` unacceptable inside an interrupt callback? Give the alternative and name the flag it relies on.
14. List the three flag families you have met so far, their peripherals, and what clears each one.

Part D - From counts to control

15. Write the loop body that reads the potentiometer every 10 ms from the L07 tick and sets the L08 LED duty. No `HAL_Delay()`.
16. `duty = (raw * 1000u) / 4095u;` — show that this cannot overflow a `uint32_t`, state what would happen with `uint16_t` intermediates, and explain why the factor is 1000 and not 999.
17. Write `servo_from_raw(uint16_t raw)` returning a pulse width clamped to the 500–2500 µs range of the lab SG90s. Give the value for `raw = 2867`.

18. The potentiometer is noisy: the last count flickers between two values and the servo buzzes. Give **two** different mitigations, one in software and one in hardware, and state a cost of each.

Part E - Sensors and evidence

19. An LDR in series with a 10 kΩ resistor between 3V3 and GND, junction to PA1. As the room gets **darker**, does `raw` rise or fall? Justify from the divider, given that an LDR's resistance rises in darkness.

20. Your night-light toggles LD2 rapidly when the light sits near the threshold. Name the phenomenon and give a fix that does not change the sensor. Then say what you would set the two thresholds to if the trip point is `raw = 2000`.

21. Fill in the evidence you would show for each claim.

Claim	SFR observation
The pin is in analog mode	
The value in `raw` is the value in the hardware	
The converter tracks the potentiometer	
The sampling time is 84 cycles	
The LDR channel was converted last	

22. A classmate's light reading never reacts to the light. Describe how the SFR view shows which channel the ADC really converts, and name the two most likely mistakes.

Exit self-assessment

☐ I can convert counts to millivolts in integer arithmetic.
☐ I can justify a sampling time from source impedance.
☐ I can calculate a conversion time and name what waits for `EOC`.
☐ I can map a reading to a duty or a servo pulse, clamped.
☐ I can diagnose threshold chatter and fix it.